

Which amino acid substitution would result in the greatest change in the net charge of a polypeptide at pH 7?

- ✓ ☒ A. Glu to Lys [75%]  
☐ B. Asp to Ala [12%]  
☐ C. Lys to Arg [5%]  
☐ D. Arg to Ser [6%]

Correct

75% answered correctly

## Explanation:

Effect of amino acid substitution on peptide net charge

| Initial amino acid | Initial charge | Final amino acid | Final charge | Difference in charge (final - initial) |
|--------------------|----------------|------------------|--------------|----------------------------------------|
|                    | -1             |                  | +1           | $1 - (-1) = +2$ ← Greatest change      |
|                    | -1             |                  | 0            | $0 - (-1) = +1$                        |
|                    | +1             |                  | +1           | $1 - (+1) = 0$                         |
|                    | +1             |                  | 0            | $0 - (+1) = -1$                        |

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The **side chain** of an amino acid may be **positively charged**, **negatively charged**, or **neutral**, depending on its **chemical structure** and the **ambient pH**. At pH 7, the side chains of lysine (Lys) and arginine (Arg) are positively charged whereas those of aspartate (Asp) and glutamate (Glu) are negatively charged. All

other amino acids have neutral side chains at pH 7.

**Removing a negatively charged** amino acid (Glu or Asp) from a polypeptide or **adding a positively charged** one (Arg or Lys) **changes the net charge** by **+1**. Conversely, **removing a positively charged** amino acid or **adding a negatively charged** one **changes the net charge** by **-1**. A Glu-to-Lys mutation removes the negative charge of Glu (change of +1) *and* adds the positive charge of Lys (change of +1), resulting in a total change of +2. All of the other substitutions listed result in a change with a magnitude of less than 2. Of the choices available, a Glu-to-Lys mutation will result in the greatest change in net charge.

**(Choice B)** An Asp-to-Ala mutation removes the negative charge of Asp (change of +1) and adds the neutral charge of Ala (change of 0) for a total change of +1.

**(Choice C)** A Lys-to-Arg mutation *removes* the positive charge of Lys (change of -1) but also *adds* the positive charge of Arg (change of +1) for a total change of 0.

**(Choice D)** An Arg-to-Ser mutation removes the positive charge of Arg (change of -1) and adds the neutral charge of Ser (change of 0) for a total change of -1.

### **Educational objective:**

Amino acid side chains may be positively charged, negatively charged, or neutral, depending on their composition and the pH of their surroundings. Removing a positively charged amino acid (Arg or Lys) or adding a negatively charged one (Glu or Asp) yields a net change in charge of -1. Adding a positively charged amino acid or removing a negatively charged one yields a net change in charge of +1.

Time spent:0 sec

Subject:

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System:

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